Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Engineering & Technology

B.Tech. | Sem-5 | Summer-2026

Course Code	:	2301CS501	**Date**	:	11-05-2026
Course Name	:	Computer Network	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	What is Computer Network? Explain Advantages and Disadvantages of Computer Network.	**4**
	(B)	Explain Star topology with advantages and disadvantages.	**3**
		OR	
		Explain Bus topology with advantages and disadvantages.	
	(C)	Discuss OSI layer with each layer functionality.	**7**
		OR	
		Compare OSI Reference model and TCP/IP protocol model.	
Q.2	**(A)**	Define role and responsibility of application layer.	**4**
	(B)	Define Web & HTTP in details.	**3**
		OR	
		Define cookies and web cache.	
	(C)	Explain SMTP with example.	**7**
		OR	
		Describe working of DNS protocol.	
Q.3	**(A)**	Discuss functions and responsibilities of transport layer.	**4**
	(B)	Explain checksum with any example.	**3**
		OR	
		Illustrate TCP slow start approach.	
	(C)	Explain stop and wait protocol.	**7**
		OR	
		Explain Go-Back-N protocol.	

Q.4	**(A)**	Discuss Network Protocols & Network Application.	**4**
	(B)	Define routing. Why routing required in network?	**3**
		OR	
		Define unicast, multicast and broadcast.	
	(C)	Describe IPv4 datagram.	**7**
		OR	
		Explain different types of IP addressing classes.	
Q.5	**(A)**	Discuss services of data link layer.	**4**
	(B)	Describe parity check technique with example.	**3**
		OR	
		Classify multiple access protocol.	
	(C)	Describe CRC technique with example.	**7**
		OR	
		Describe bit stuffing and byte stuffing with example.	

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